

Supplemental Material

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Supplementary Appendix 1. Data analysis overview, analytic notes for individual cohorts, and further details on covariate and outcome definitions and ascertainment

1.1 Overview:

In order to determine the cohorts eligible for this paper, in February/March 2023 all United States-based cohorts in CKD-PC with cardiovascular outcome data available were contacted regarding their interest in this topic and to confirm availability of data. 48 cohorts were contacted, one was not able to participate due to a lack of all-cause mortality data and one was not able to send de-identified data in time for analyses, resulting in 46 cohorts in this analysis. Cohorts were required to have at least 5 years of follow-up for the 95th percentile of time to events [cases].

As previously described,⁵¹ the collaborating cohorts were asked to compile a dataset with approximately 25 variables (Required: main exposure [age, sex, race/ethnicity, history of CVD, smoking, diabetes, systolic blood pressure, antihypertensive medications, total cholesterol, HDL cholesterol, statin use, BMI, eGFR], outcome [fatal and non-fatal myocardial infarction, fatal and non-fatal ischemic and hemorrhagic stroke, fatal and non-fatal heart failure hospitalization, all-cause mortality]; In addition data were captured if available on: social deprivation index, albuminuria, and hemoglobin A1c). To be consistent across cohorts, the CKD-PC Data Coordinating Center sent definitions for those variables to participating cohorts. We instructed studies not to impute any variables.

The CKD-PC data request and processing procedures are as follows. After obtaining opt-in preferences from cohorts for the topics for each phase, the Data Coordinating Center (DCC) requests de-identified data using a specific data request document describing the variables and preferred definitions needed for the current phase of the CKD-PC. Cohorts work with the DCC on any data use agreements, IRB approvals, and other logistic issues for de-identified data transfer. The DCC also advises on any differences in definitions or questions on data formatting. Cohorts then provide de-identified data (in whatever program format, e.g., Stata, SAS, csv) via a secure data transfer provided or their own secure transfer program/platform. Data is stored on a secure password protected network server that is only accessed by limited faculty and staff (<10). All those faculty and staff have completed HIPAA and CITI certification and have signed internal data use agreements to not use the data for any other than stated purposed and to not remove the data from that network drive. The CKD-PC does not share data with any external parties. Once data is received and stored in the network drive, the DCC programmer reviews the data and the data dictionary provided by the cohort to check for any missing information, outliers, and potentials issues with variable units, dates, etc. Any questions are sent to the cohort representatives for data checking and cleaning. Further data checking is done throughout the analysis process for each CKD-PC paper, including a review from a cohort representative of all tables and figures to confirm their cohort representation.

For the 46 cohorts in this specific study, the DCC at Johns Hopkins University conducted the analysis. Then the DCC meta-analyzed the estimates across cohorts using Stata.

As detailed in our previous reports,³⁸ each cohort was instructed to standardize their serum creatinine and report its method when available. The reported creatinine standardization allows grouping studies into studies that reported using a standard IDMS traceable method or conducted some serum creatinine standardization to IDMS traceable methods (ARIC, CARDIA, CRIC, Geisinger, JHS, Rancho Bernardo, REGARDS) and studies where the creatinine standardization was not done (CHS, Framingham, MESA). For those cohorts without standardization, the creatinine levels were reduced by 5%, the calibration factor used to adjust non-standardized MDRD Study samples to IDMS.^{37, 40} We did not adjust creatinine levels in those studies with unknown standardization status (OLDW all cohorts).

Table S18. eGFR calculated using the 2021 CKD-EPI creatinine³⁹

Sex	Serum Creatinine (mg/dL)	Equation
CKD-EPI creatinine equation		
Female	≤0.7	$GFR = 142 \times (Scr/0.7)^{-0.241} \times 0.9938^{Age}$
	>0.7	$GFR = 142 \times (Scr/0.7)^{-1.200} \times 0.9938^{Age}$
Male	≤0.9	$GFR = 142 \times (Scr/0.9)^{-0.302} \times 0.9938^{Age}$
	>0.9	$GFR = 142 \times (Scr/0.9)^{-1.200} \times 0.9938^{Age}$

The selection of knots for eGFR and urine albumin-to-creatinine ratio was based on clinical thresholds.⁴¹

1.2 Details of the analysis:

Survival analysis was conducted stratified by sex on the age scale allowing for left censoring at the age of entry and right censoring at the age of death or exit from follow-up. For each outcome (CVD, ASCVD, HF, CHD and for use in competing risks models death prior to each of these) we conducted a separate Cox regression.

Meta analysis of hazard ratios from all development cohorts was conducted using univariate random effects models. The meta-analyzed hazard ratio was applied to all development models in a Cox model to obtain the adjusted baseline cumulative hazard (centered at age 55, non-HDL-C 3.5, HDL-C 1.3, SBP 130 and absence of other risk factors). This baseline log cumulative hazard was smoothed using a cubic spline with knots at 35, 55, 65, 75 and 85 of age (the approximate 1, 25, 50, 75 and 99 percentiles). For each outcome, age specific log hazards were calculated and the median at each age and sex specific group were calculated. The overall result was then smoothed using a linear age.

Details on the meta-analysis method are as follows: All analyses were performed separately in each cohort. Next, estimates were pooled using random-effects meta-analysis using the Stata “metan” command (Stata version 16, College Station, TX). This two-stage approach allows close evaluation of individual study results to ensure no errors. The random-effects meta-analysis allows for estimation of both within study random variation and between study systematic variation (heterogeneity; International Journal of Epidemiology 2010;39:1345–1359). Given the complexity and number of the competing risk models and the size of the datasets, we employ a univariate meta-analysis rather than a multivariate approach (e.g., “mvmeta”), which requires prohibitively long computing time. In similar analyses, “metan” and “mvmeta” had <2% difference in estimates. We routinely share statistical code upon request. The source data were released to us using specific data use agreements so we cannot share the data themselves but we reply to queries including running analyses to check on the results using the full datasets.

Absolute risk of each CVD outcome was calculated using a competing risks model of the cause specific hazard of the CVD outcome and its paired mortality (free of this type of CVD). This model used the meta-analyzed hazard ratios and median baseline hazard.

Discrimination was calculated using time-dependent Harrel’s C-statistic. Calibration was estimated by graphing the observed risk within each decile of predicted risk. The slope was calculated as a summary measure of calibration with a slope of 1.0 indicating good calibration (<1 indicating over-prediction and >1 indicating under-prediction). Deciles were combined if they had <5 events. Calibration was examined overall and within subgroups. Improvement in calibration between the base model and larger models were tested based on a Wilcoxon matched pair rank-sum test of the paired comparisons between the absolute value of the difference between the slope and 1.0 in the two models being compared. NRI was calculated using the event rate method which uses the event rates as the cutoff to calculate sensitivity and specificity for each model and the difference in each is summed to obtain the NRI.⁶²

Calibration figures applied minimum criteria to focus on reliable data. Specifically, the follow-up period was shortened from 10 years to 8 or 5 years if the 95th percentile for events was 8-10 years or 5-8 years respectively. In addition, deciles were merged when there were fewer than 5 events in the decile. Cohorts with fewer than 10 events for the specific outcome, subgroup and follow-up time were not plotted.

Regression approximations were calculated for a more rapid calculation of the competing risks models. For each model, the logit (log (risk/1-risk)) absolute risk was fit using linear regression on the same predictors as in the relative hazard model plus age. Approximations were considered adequate since their r-square was greater than 0.99 and visual inspection of full model vs. approximate model absolute risk calculations indicated very tight agreement at risks below 10% and excellent agreement at higher risks.

1.3 Notes for individual cohorts:

ARIC: Visit 4 was used as baseline, which was conducted between 1996-98. Hemoglobin A1c was not available in this study.

CARDIA: Year 10 was used as baseline, which was conducted between 1995-96. Hemoglobin A1c was not available in this study.

CHS: Visit 5 was used as baseline, which was conducted between 1995-98. This cohort had urine albumin-to-creatinine ratio measures at a different visit of HDL cholesterol, thus not included in the Base+albuminuria analysis. Hemoglobin A1c was not available in this study.

CRIC: Visit 1 was used as baseline, which was conducted between 2003-07. Of the ACR data, 25% were estimated from PCR. Hemoglobin A1c was not available in this study.

Framingham: Visit 1 was used as baseline, which was conducted between 1995-98. Hemoglobin A1c was not available in this study.

Geisinger: Baseline was defined as first outpatient visit with both serum creatinine and systolic blood pressure measurements after 1/1/2009, as well as enrolled in the system for at least one year. Albumin to creatinine ratio was abstracted from the closest lab measurement within 3 years prior to the baseline. When albuminuria data were not available, the closest proteinuria (protein to creatinine ratio, PCR, or dipstick) within 3 years prior to the baseline were used to estimate ACR using a validated algorithm.⁴² Of the ACR data 4.3% were estimated from PCR and 53% from dipstick.

JHS: Visit 1 was used as baseline, which was conducted between 2000-04. History of HF was not available in this study. JHS participants that were also in ARIC were excluded.

MESA: Visit 1 was used as baseline, which was conducted between 2000-02. Hemoglobin A1c was not available at baseline in this study.

OLDW: This study used de-identified electronic health record (EHR) data from the Optum Labs Data Warehouse (OLDW). The database contains longitudinal health information on enrollees and patients, representing a mixture of ages and geographical regions across the United States. The EHR-derived data includes a subset of EHR data that has been normalized and standardized into a single database.²⁹ Cohort inclusion criteria was more than 50 events of ASCVD after excluding missing values of main exposure variables and more than 5-year of 95th percentile of follow-up in events. One OLDW cohort with >70% missing BMI data was excluded. Baseline was considered first serum creatinine after 2008 at a visit with a blood pressure measurement and in the health system for at least a year. Smoking status might be under measured in this study. We validated separately in each cohort in this study. No cause of death information available. ASCVD was defined by ICD codes from encounters.

Social deprivation index (SDI) during 2015-2020 was linked to the zip code data at the individual participant level by OLDW staff. SDI was rounded to the nearest decile from 1 to 10. The year used for assigning the zip code for SDI was the closest year to the baseline for each participant and excluded if >5 years before or >3 years after the patient's baseline year. SDI missing value indicator identified patients whose zip code of residence was missing.

Albumin to creatinine ratio was abstracted from the closest lab measurement within 3 years prior to the baseline. When albuminuria data were not available, the closest proteinuria (protein to creatinine ratio, PCR, or dipstick) within 3 years prior to the baseline were used to estimate ACR using a validated algorithm.⁴² Of the ACR data 0.81% were estimated from PCR and 79% from dipstick.

Rancho Bernardo: Visit 1 was used as baseline, which was conducted between 1992-97. Hemoglobin A1c was not available in this study.

REGARDS: The initial in-home visit was used as baseline, which was conducted between 2003-07. History of suspected HF was not used in the analyses.

1.4 Numbers of missing covariates, N (%)

Our consortium includes both research cohorts and clinical databases. The main mechanisms of missingness seems somewhat different in these two cohort types. Specifically, in the former, a set of variables are evaluated based on a prespecified protocol, and the missingness is often due to lack of specimens, difficulty of measurement in specific participants, or lack of consent for specific tests. Thus, typically, individuals with missing data are less healthy than those without missing data. In the latter, data are collected according to clinical routine or indication. Thus, missing data may be seen in healthier populations. Nonetheless, we first extracted individuals with available data of variables required for analyses (i.e., age, sex, race/ethnicity, history of CVD, smoking, diabetes, systolic blood pressure, antihypertensive medications, total cholesterol, HDL cholesterol, statin use and body mass index) as well as eGFR given overall wide availability in our datasets. The display below summarizes the missing rate of race, albuminuria (either by urine albumin-to-creatinine ratio, urine protein-to-creatinine ratio, or urine dipstick), and hemoglobin A1c (HbA1c) used in this study by each cohort.

Table S19. Percent missing

Development datasets	N	Race/ethnicity	ACR	HbA1c
ARIC	8,838	0%	0.79%	100%
CARDIA	2,738	0%	10%	100%
CHS	2,543	0%	100%	100%
Framingham	2,611	0%	0%	100%
Geisinger	166,219	0%	47%	72%
JHS	2,399	0%	32%	1.7%
MESA	5,928	0%	<0.5%	100%
OLDW 1	151,346	0%	53%	65%
OLDW 3	117,111	0%	61%	77%
OLDW 4	50,694	0%	59%	76%
OLDW 8	640,772	0%	54%	72%
OLDW 10	222,974	0%	35%	76%
OLDW 12	53,835	0%	53%	35%
OLDW 19	172,519	0%	51%	76%
OLDW 20	227,092	0%	55%	49%
OLDW 21	443,351	0%	54%	71%
OLDW 22	109,510	0%	59%	69%
OLDW 23	288,625	0%	72%	78%
OLDW 25	141,928	0%	57%	66%
OLDW 26	54,292	0%	53%	56%
OLDW 27	85,044	0%	62%	68%
OLDW 28	57,631	0%	68%	76%
OLDW 29	62,785	0%	69%	63%
OLDW 32	101,430	0%	51%	63%
OLDW 33	109,704	0%	64%	52%
Validation Datasets				
CRIC	1,948	0%	3.4%	100%
OLDW 2	10,356	0%	54%	81%
OLDW 5	70,669	0%	45%	78%
OLDW 6	33,214	0%	60%	71%
OLDW 7	132,583	0%	21%	55%
OLDW 9	569,383	0%	64%	73%
OLDW 11	332,600	0%	46%	74%
OLDW 13	82,367	0%	28%	66%
OLDW 14	112,487	0%	55%	66%
OLDW 15	285,469	0%	61%	76%
OLDW 16	82,144	0%	55%	73%

OLDW 17	229,733	0%	39%	38%
OLDW 18	37,726	0%	46%	56%
OLDW 24	114,453	0%	67%	63%
OLDW 30	96,943	0%	59%	69%
OLDW 31	70,521	0%	60%	42%
OLDW 34	185,799	0%	67%	64%
OLDW 35	101,997	0%	82%	57%
Rancho Bernardo	1,053	14%	<0.5%	100%
REGARDS	17,980	0%	3.6%	79%

1.5 Covariate assessment

1.5.1 Table S20. Race/Ethnicity ascertainment in each cohort

Cohort	Race/ethnicity groups identified in cohort	Ascertainment and notes	Reference
ARIC	Question: “Now I would like to know the race or ethnic identification, sex, and marital status of each household member.” Variable: RACEGRP Asian Black American Indian or Alaskan Indian White Eligibility form 6b. Hispanic Yes No	The ARIC Study collects data in four diverse communities. This design was chosen so that data could be obtained for groups which differ by geography, race, and socio-economic status. The ARIC study was not designed to select a random or representative sample of the entire U.S. population. Race was self-reported in ARIC and includes more race groups than Black and White, but there are very few of those other groups (34 self-reporting as Asian and 14 as American Indian or Alaskan Indian). Race in ARIC is also confounded by field center as most of the Black participants are from the Jackson, Mississippi center.	Household enumeration form: https://sites.csc.unc.edu/aric/sites/default/files/public/forms/HEFA.pdf
CARDIA	Variable: A01RACE1 Hispanic Black, not Hispanic White, not Hispanic American Indian or Alaskan native Asian or Pacific Islander Don’t know No answer	Self-report on telephone interview for selection and monitoring. CARDIA then recruited individuals self-reporting as White or Black only.	https://www.cardia.dopm.uab.edu/images/more/pdf/D10001.PDF
CHS	Variable: RACE01 White Black American/Indian/Alaskan Native Asian/Pacific Islander Other (specify ROTHER01) Variable: HISP01 Do you consider yourself of Hispanic origin? Yes No	Self-report on the CHS eligibility form.	CHS eligibility form: https://chs-nhlbi.org/forms/r1p1

Cohort	Race/ethnicity groups identified in cohort	Ascertainment and notes	Reference
CRIC	Q: Using the categories below, what do you consider to be your racial background? (check yes/no) American Indian/Alaskan Native Asian/Asian American Black/African American Native Hawaiian/Other Pacific Islander White/Caucasian Q: If Asian/Asian American, do you consider yourself to be (check all that apply) Chinese East Indian/South Asian Japanese Filipino Korean Southeast Asian Other Don't know Q: If Black or African American, do you consider yourself to be (check all that apply) American African Haitian Jamaican Cuban Puerto Rican Dominican Other Caribbean Island Central/South American Other Don't know	Race was Self-Identified in Demographic Information Form	Case Form: https://repository.niddk.nih.gov/media/studies/cric/forms/demographic_information.pdf

Cohort	Race/ethnicity groups identified in cohort	Ascertainment and notes	Reference
	If you checked Hispanic/Latino, do you consider yourself to be (check all that apply) Mexican American/Mexican Central American South American Puerto Rican Cuban Dominican Spaniard or Portuguese Other Don't Know		
Framingham	Not systematically collected until later Framingham Offspring exams	The population is considered to be predominantly White.	https://www.bu.edu/fhs/share/annotated_forms/vr_raceall_2011_a_1257s_annotated_form.pdf
Geisinger	Race: American Indian Or Alaska Native Asian Black Or African American Native Hawaiian Or Other Pacific Islander Other Unknown White Ethnicity: Hispanic or Latino Not Hispanic or Latino Unknown	As reported in the electronic health record.	
JHS	All participants are Black	Recruited African American participants in the Jackson, Mississippi Metropolitan Statistical Area. Individuals approached included ARIC participants at the Jackson site (excluded from this analysis so there is no overlap), a random sample, volunteers, and a family study including participants in any of the prior 3 groups with a family size meeting eligibility requirement (Ethn Dis . 2005 Autumn;15(4 Suppl 6):S6-18-29).	

Cohort	Race/ethnicity groups identified in cohort	Ascertainment and notes	Reference
MESA	Q: Are you Spanish/Hispanic/Latino? If yes: Mexican, Chicano, Mexican-American Dominican Puerto Rican Cuban Q: Which of the following best describes you? African-American or Black Asian Chinese Filipino Japanese Korean Vietnamese Asian Indian Caucasian or White Native Hawaiian or other Pacific Islander Guamanian or Chamorro Samoan Micronesia Tahitian American Indian or Alaska Native Did not identify	As part of the study design, approximately 38% of the recruited participants are white, 28% African-American, 22% Hispanic, and 12% Asian, predominantly of Chinese descent. Race was identified as self-report on the Screening/Recruitment form. Separate questions for Hispanic other groups, and if respondent names a Spanish/Hispanic/Latino group instead of the race classification question, wording for interviewers to state that “Some Hispanics also identify themselves with one of these other groups”	Screening/Recruitment form: https://www.mesa-nhlbi.org/PublicDocs/010101-011231/MESABaselineExamForms/screener.pdf. Accessed 2021
OLDW cohorts	Race: Africa American, Asian, Caucasian, Other/Unknown Ethnicity: Hispanic, Not Hispanic, Unknown	As reported in the electronic health record.	
Rancho Bernardo	Collected upon entry into the study. Asked as White: Yes/No	Self-report. Because 99.4% of the population are White and not Hispanic, race was denoted as “white: yes/no”.	

Cohort	Race/ethnicity groups identified in cohort	Ascertainment and notes	Reference
REGARDS	From recruitment: White, non-Hispanic Black, non-Hispanic	Self-reported race/ethnicity as part of the recruitment. Interviewers ask QLatino – “Are you Hispanic or Latino?” And those responding Yes are not eligible for the study. Note from Baseline_CATI_Codebook_REGARDS08_17_05: “Beginning 6/9/04, if respondent is Latino, return to QAgess question and select another potential respondent from household and proceed. If there are no other age eligible respondents, thank and terminate.” Self report question for baseline: QRace: “What is your race? Would you say White, Black or African American, Asian, Native Hawaiian or Other Pacific Islander American Indian, Alaska Native or some other race?” Those that did not identify as Black or White are then directed to Sorry2: “Thank you for taking a few minutes to speak with me, but for the purposes of this study we are trying to interview equal numbers of different racial and ethnic groups and have already met our quota in your area. Thank you again, and have a great day.”	Computer assisted telephone interview codebook documentation: https://www.uab.edu/soph/regardsstudy/images/documents/Baseline_CATI_Codebook_REGARDS08_17_05.doc

1.5.2 Table S21. Other medical history variables

To be consistent across cohorts, the CKD-PC Data Coordinating Center sent definitions for those medical history variables to participating cohorts (outlined below). We instructed studies not to impute any variables and let us know of any differences in definitions.

Medical history variable	Definition
Total cholesterol	Total cholesterol (mmol/L)
HDL-Cholesterol	HDL cholesterol (mmol/L)
Systolic blood pressure	Systolic blood pressure (mmHg)
Diabetes mellitus	Glycated hemoglobin A1c $\geq 6.5\%$ or fasting glucose ≥ 7.0 mmol/L (≥ 126 mg/dL) or non-fasting glucose ≥ 11.1 mmol/L (≥ 200 mg/dL) or use of glucose lowering drugs (ADA 2010 criteria). Self-report of physician diagnosed diabetes can be included. Identification at any time during study period (from baseline to follow-up).
Hemoglobin A1c (HbA1c)	HbA1C (glycated hemoglobin) (%) measured by a method certified by the National Glycohaemoglobin Standardisation Program (NGSP) and standardized to the Diabetes Control and Complications Trial (DCCT)
Smoking	Information on current-, former-, and never-smoker
Height	Height (m)
Body weight	Body weight (kg)
Medications for hypertension	Taking any medications prescribed for lowering blood pressure. In administrative data we used ATC codes: C02, C03, C07, C08, C09
Statin use	Statin use; In administrative data ATC code C10AA, C10B
Urine albumin-to-creatinine ratio (ACR)	Urinary albumin (mg) / Urinary creatinine (g)
Urine protein-to-creatinine ratio (PCR)	Urinary protein (mg) / Urinary creatinine (g)
Dipstick proteinuria	Dipstick proteinuria
Serum creatinine	Serum creatinine standardized to isotope dilution mass spectrometry (IDMS) (mg/dL) -- specify assay and standardization method

1.6 Outcome definitions and ascertainment

1.6.1 Table S22. ICD codes used to define outcome if not specified

Outcome	ICD-9 codes	ICD-10 codes
Myocardial infarction (MI)	410	I21, I22
Hemorrhagic and Ischemic stroke	431, 432, 433.x1, 434.x1	I61, I62, I63
Heart failure	428	I50

1.6.2 Table S23. Details of individual cohorts

Development datasets	Atherosclerotic cardiovascular disease ascertainment type	Heart Failure	All-cause mortality ascertainment
ARIC	- CHD: fatal MI and non-fatal MI adjudicated by a physician panel - Stroke: Ischemic and hemorrhagic stroke adjudicated by a physician panel; fatal stroke by death with an incident stroke	ICD codes from hospital discharge or death registry, adjudicated after 2005	Active and linkage to a registry
CARDIA	- CHD: fatal CHD and non-fatal MI adjudicated by a physician panel - Stroke: fatal, ischemic and hemorrhagic stroke adjudicated by a physician panel	Adjudicated by a physician panel	Linkage
CHS	Atherosclerotic CHD, cerebrovascular disease, and other cardiovascular disease from cause of death adjudicated by a physician panel	Adjudicated by CHS Events Subcommittee	Active
Framingham	- CHD: fatal MI and non-fatal MI by medical chart review - Stroke: fatal, ischemic and hemorrhagic stroke by medical chart review	Not hospitalized, diagnosed on basis of on exam or MD notes or hospitalized heart failure	Active (with medical chart review)
Geisinger	- CHD: MI by ICD codes from hospital discharge - Stroke: ischemic and hemorrhagic stroke by ICD codes from hospital discharge	ICD codes from hospital discharge	Linkage
JHS	- CHD: fatal CHD and MI adjudicated by a physician panel - Stroke: fatal, ischemic and hemorrhagic stroke adjudicated by a physician panel	ICD codes from hospital discharge or death registry, adjudicated after 2005	Active (with medical chart review)

MESA	• CHD: fatal MI and non-fatal MI adjudicated by a physician panel • Stroke: fatal, ischemic and hemorrhagic stroke defined as new onset of neurologic symptoms last 24 hours or leading to death with a clinically relevant lesion on brain imaging not caused by a nonvascular cause, adjudicated by a physician panel	Reviewers classified CHF as definite, probable, or absent. Definite or probable CHF required heart failure symptoms, such as shortness of breath or edema, as asymptomatic disease is not a MESA endpoint. In addition to symptoms, probable CHF required CHF diagnosed by a physician and patient receiving medical treatment for CHF. Definite CHF required one or more other criteria, such as pulmonary edema/congestion by chest X-ray; dilated ventricle or poor LV function by echocardiography or ventriculography; or evidence of left ventricular diastolic dysfunction. We considered participants not meeting any criteria, including just a physician diagnosis of CHF without any other evidence, as having no CHF.	Active
OLDW derivation cohorts	• CHD: MI by ICD codes from hospital discharge • Stroke: ischemic and hemorrhagic stroke by ICD codes from hospital discharge	ICD codes from hospital discharge	Medicare/Medicaid claims has linkage and commercial claims has hospital discharge
Validation Datasets			
CRIC	• CHD: fatal MI and MI adjudicated by a physician panel • Stroke: fatal, ischemic and hemorrhagic stroke adjudicated by a physician panel	Heart failure events were adjudicated by 2 independent physicians who reviewed hospitalization records, and any differences were resolved by discussion.	Active and linkage to a registry
OLDW validation cohorts	• CHD: MI by ICD codes from hospital discharge	ICD codes from hospital discharge	Medicare/Medicaid claims has linkage and commercial claims has hospital discharge

	Stroke: ischemic and hemorrhagic stroke by ICD codes from hospital discharge		
Rancho Bernardo	- CHD: fatal MI by ICD codes from death registry; MI by self-report - Stroke: fatal stroke by ICD codes from death registry; ischemic and hemorrhagic stroke by self-report	Self-report and fatal events using ICD codes	Linkage to death registry
REGARDS	- CHD: fatal MI and MI adjudicated by a physician panel - Stroke: fatal, ischemic and hemorrhagic stroke adjudicated by a physician panel	Heart failure events and heart failure deaths were adjudicated by 2 independent clinical investigators who reviewed hospitalization records, and any differences were resolved by discussion.	Active

Supplementary Appendix 2. Acronyms or abbreviations for cohorts included in the current study and their key references linked to the Web references

ARIC	Atherosclerosis Risk in Communities Study ²²
CARDIA	Coronary Artery Risk Development in Young Adults ²³
CHS	Cardiovascular Health Study ²⁴
CRIC	Chronic Renal Insufficiency Cohort Study ³¹
Framingham	Framingham Heart Study ²⁵
Geisinger	Geisinger Health System ²⁸
JHS	Jackson Heart Study ²⁶
MESA	Multi-Ethnic Study of Atherosclerosis ²⁷
OLDW	Optum Labs Data Warehouse ²⁹
Rancho Bernardo	Rancho Bernardo Study ³²
REGARDS	Reasons for Geographic And Racial Differences in Stroke Study ³⁰

Supplementary Appendix 3. Acknowledgements and funding for collaborating cohorts

Study	List of sponsors
ARIC	The Atherosclerosis Risk in Communities study has been funded in whole or in part with Federal funds from the National Heart, Lung, and Blood Institute, National Institutes of Health, Department of Health and Human Services, under Contract nos. (HHSN268201700001I, HHSN268201700002I, HHSN268201700003I, HHSN268201700004I, HHSN268201700005I). The authors thank the staff and participants of the ARIC study for their important contributions.
CARDIA	The Coronary Artery Risk Development in Young Adults Study (CARDIA) is conducted and supported by the National Heart, Lung, and Blood Institute (NHLBI) in collaboration with the University of Alabama at Birmingham (HHSN268201800005I & HHSN268201800007I), Northwestern University (HHSN268201800003I), University of Minnesota (HHSN268201800006I), and Kaiser Foundation Research Institute (HHSN268201800004I). This manuscript has been reviewed by CARDIA for scientific content.
CHS	This research was supported by contracts HHSN268201200036C, HHSN268200800007C, HHSN268201800001C, N01HC55222, N01HC85079, N01HC85080, N01HC85081, N01HC85082, N01HC85083, N01HC85086, 7592021D00006, and grants U01HL080295 and U01HL130114 from the National Heart, Lung, and Blood Institute (NHLBI), with additional contribution from the National Institute of Neurological Disorders and Stroke (NINDS). Additional support was provided by R01AG023629 from the National Institute on Aging (NIA). A full list of principal CHS investigators and institutions can be found at CHS-NHLBI.org.
CRIC	Funding for the CRIC Study was obtained under a cooperative agreement from National Institute of Diabetes and Digestive and Kidney Diseases (U01DK060990, U01DK060984, U01DK061022, U01DK061021, U01DK061028, U01DK060980, U01DK060963, and U01DK060902). In addition, this work was supported in part by: the Perelman School of Medicine at the University of Pennsylvania Clinical and Translational Science Award NIH/NCATS UL1TR000003, Johns Hopkins University UL1 TR-000424, University of Maryland GCRC M01 RR-16500, Clinical and Translational Science Collaborative of Cleveland, UL1TR000439 from the National Center for V 2018.10.30 Advancing Translational Sciences (NCATS) component of the National Institutes of Health and NIH roadmap for Medical Research, Michigan Institute for Clinical and Health Research (MICHHR) UL1TR000433, University of Illinois at Chicago CTSA UL1RR029879, Tulane COBRE for Clinical and Translational Research in Cardiometabolic Diseases P20 GM109036, Kaiser Permanente NIH/NCRR UCSF-CTSI UL1 RR-024131.
Framingham	NHLBI Framingham Heart Study (N01-HC-25195).
Geisinger	Geisinger Clinic; NIDDK R01DK100446
JHS	The Jackson Heart Study (JHS) is supported and conducted in collaboration with Jackson State University (HHSN268201800013I), Tougaloo College (HHSN268201800014I), the Mississippi State Department of Health (HHSN268201800015I) and the University of Mississippi Medical Center (HHSN268201800010I, HHSN268201800011I and HHSN268201800012I) contracts from the National Heart, Lung, and Blood Institute (NHLBI) and the National Institute for Minority Health and Health Disparities (NIMHD). The authors also wish to thank the staffs and participants of the JHS.
MESA	This research was supported by contracts HHSN268201500003I, N01-HC-95159, N01-HC-95160, N01-HC-95161, N01-HC-95162, N01-HC-95163, N01-HC-95164, N01-HC-95165, N01-HC-95166, N01-HC-95167, N01-HC-95168 and N01-HC-95169 from the National Heart, Lung, and Blood Institute and by grants UL1-TR-000040 and UL1-TR-001079 from NCRR. The authors thank the other investigators, the staff, and the

	participants of the MESA study for their valuable contributions. A full list of participating MESA investigators and institutions can be found at http://www.mesa-nhlbi.org .
OLDW	
Rancho Bernardo	NIA AG07181 and AG028507 NIDDK DK31801
REGARDS	This research project is supported by cooperative agreement U01 NS041588 co-funded by the National Institute of Neurological Disorders and Stroke (NINDS) and the National Institute on Aging (NIA), National Institutes of Health, Department of Health and Human Service. The content is solely the responsibility of the authors and does not necessarily represent the official views of the NINDS or the NIA. The authors thank the other investigators, the staff, and the participants of the REGARDS study for their valuable contributions. A full list of participating REGARDS investigators and institutions can be found at: https://www.uab.edu/soph/regardsstudy/ Additional funding was provided by National Heart, Lung, and Blood Institute (NHLBI) grant R01 HL080477. Representatives from NHLBI did not have any role in the design and conduct of the study, the collection, management, analysis, and interpretation of the data, or the preparation or approval of the manuscript.

Supplementary Appendix 4. Detailed example applying the simplified PREVENT models

Table S12 provides the coefficients for all 10 year and 30 year CVD models. Below we display the equations.

Table S24. 10 year Equations

10 year Total CVD Risk	
Women	$\text{log-Odds} = -3.307728 + 0.7939329 \times (\text{age} - 55) / 10 + 0.0305239 \times (\text{Tc} - \text{HDL} - 3.5) - 0.1606857 \times (\text{HDL} - 1.3) / 0.3 - 0.2394003 \times (\min(\text{SBP}, 110) - 110) / 20 + 0.360078 \times (\max(\text{SBP}, 110) - 130) / 20 + 0.8667604 \times (\text{if diabetes}) + 0.5360739 \times (\text{if current smoker}) + 0.6045917 \times (\min(\text{eGFR}, 60) - 60) / -15 + 0.0433769 \times (\max(\text{eGFR}, 60) - 90) / -15 + 0.3151672 \times (\text{if using anti-hypertensive medication}) - 0.1477655 \times (\text{if using statin}) - 0.0663612 \times (\text{if using anti-hypertensive medication}) \times (\max(\text{SBP}, 110) - 130) / 20 + 0.1197879 \times (\text{if using statin}) \times (\text{Tc} - \text{HDL} - 3.5) - 0.0819715 \times (\text{age} - 55) / 10 \times (\text{Tc} - \text{HDL} - 3.5) + 0.0306769 \times (\text{age} - 55) / 10 \times (\text{HDL} - 1.3) / 0.3 - 0.0946348 \times (\text{age} - 55) / 10 \times (\max(\text{SBP}, 110) - 130) / 20 - 0.27057 \times (\text{age} - 55) / 10 \times (\text{if diabetes}) - 0.078715 \times (\text{age} - 55) / 10 \times (\text{if current smoker}) - 0.1637806 \times (\text{age} - 55) / 10 \times (\min(\text{eGFR}, 60) - 60) / -15$
	$\text{Risk} = \exp(\text{log-Odds}) / (1 + \exp(\text{log-Odds}))$
Men	$\text{log-Odds} = -3.031168 + 0.7688528 \times (\text{age} - 55) / 10 + 0.0736174 \times (\text{Tc} - \text{HDL} - 3.5) - 0.0954431 \times (\text{HDL} - 1.3) / 0.3 - 0.4347345 \times (\min(\text{SBP}, 110) - 110) / 20 + 0.3362658 \times (\max(\text{SBP}, 110) - 130) / 20 + 0.7692857 \times (\text{if diabetes}) + 0.4386871 \times (\text{if current smoker}) + 0.5378979 \times (\min(\text{eGFR}, 60) - 60) / -15 + 0.0164827 \times (\max(\text{eGFR}, 60) - 90) / -15 + 0.288879 \times (\text{if using anti-hypertensive medication}) - 0.1337349 \times (\text{if using statin}) - 0.0475924 \times (\text{if using anti-hypertensive medication}) \times (\max(\text{SBP}, 110) - 130) / 20 + 0.150273 \times (\text{if using statin}) \times (\text{Tc} - \text{HDL} - 3.5) - 0.0517874 \times (\text{age} - 55) / 10 \times (\text{Tc} - \text{HDL} - 3.5) + 0.0191169 \times (\text{age} - 55) / 10 \times (\text{HDL} - 1.3) / 0.3 - 0.1049477 \times (\text{age} - 55) / 10 \times (\max(\text{SBP}, 110) - 130) / 20 - 0.2251948 \times (\text{age} - 55) / 10 \times (\text{if diabetes}) - 0.0895067 \times (\text{age} - 55) / 10 \times (\text{if current smoker}) - 0.1543702 \times (\text{age} - 55) / 10 \times (\min(\text{eGFR}, 60) - 60) / -15$
	$\text{Risk} = \exp(\text{log-Odds}) / (1 + \exp(\text{log-Odds}))$
10 year ASCVD Risk	
Women	$\text{log-Odds} = -3.819975 + 0.719883 \times (\text{age} - 55) / 10 + 0.1176967 \times (\text{Tc} - \text{HDL} - 3.5) - 0.151185 \times (\text{HDL} - 1.3) / 0.3 - 0.0835358 \times (\min(\text{SBP}, 110) - 110) / 20 + 0.3592852 \times (\max(\text{SBP}, 110) - 130) / 20 + 0.8348585 \times (\text{if diabetes}) + 0.4831078 \times (\text{if current smoker}) + 0.4864619 \times (\min(\text{eGFR}, 60) - 60) / -15 + 0.0397779 \times (\max(\text{eGFR}, 60) - 90) / -15 + 0.2265309 \times (\text{if using anti-hypertensive medication}) - 0.0592374 \times (\text{if using statin}) - 0.0395762 \times (\text{if using anti-hypertensive medication}) \times (\max(\text{SBP}, 110) - 130) / 20 + 0.0844423 \times (\text{if using statin}) \times (\text{Tc} - \text{HDL} - 3.5) - 0.0567839 \times (\text{age} - 55) / 10 \times (\text{Tc} - \text{HDL} - 3.5) + 0.0325692 \times (\text{age} - 55) / 10 \times (\text{HDL} - 1.3) / 0.3 - 0.1035985 \times (\text{age} - 55) / 10 \times (\max(\text{SBP}, 110) - 130) / 20 - 0.2417542 \times (\text{age} - 55) / 10 \times (\text{if diabetes}) - 0.0791142 \times (\text{age} - 55) / 10 \times (\text{if current smoker}) - 0.1671492 \times (\text{age} - 55) / 10 \times (\min(\text{eGFR}, 60) - 60) / -15$
	$\text{Risk} = \exp(\text{log-Odds}) / (1 + \exp(\text{log-Odds}))$
Men	$\text{log-Odds} = -3.500655 + 0.7099847 \times (\text{age} - 55) / 10 + 0.1658663 \times (\text{Tc} - \text{HDL} - 3.5) - 0.1144285 \times (\text{HDL} - 1.3) / 0.3 - 0.2837212 \times (\min(\text{SBP}, 110) - 110) / 20 + 0.3239977 \times (\max(\text{SBP}, 110) - 130) / 20 + 0.7189597 \times (\text{if diabetes}) + 0.3956973 \times (\text{if current smoker}) + 0.3690075 \times (\min(\text{eGFR}, 60) - 60) / -15 + 0.0203619 \times (\max(\text{eGFR}, 60) - 90) / -15 + 0.2036522 \times (\text{if using anti-hypertensive medication}) - 0.0865581 \times (\text{if using statin}) - 0.0322916 \times (\text{if using anti-hypertensive medication}) \times (\max(\text{SBP}, 110) - 130) / 20 + 0.114563 \times (\text{if using statin}) \times (\text{Tc} - \text{HDL} - 3.5) - 0.0300005 \times (\text{age} - 55) / 10 \times (\text{Tc} - \text{HDL} - 3.5) + 0.0232747 \times (\text{age} - 55) / 10 \times (\text{HDL} - 1.3) / 0.3 - 0.0927024 \times (\text{age} - 55) / 10 \times (\max(\text{SBP}, 110) - 130) / 20 - 0.2018525 \times (\text{age} - 55) / 10 \times (\text{if diabetes}) - 0.0970527 \times (\text{age} - 55) / 10 \times (\text{if current smoker}) - 0.1217081$

$$\times (\text{age} - 55) / 10 \times (\text{min}(\text{eGFR}, 60) - 60) / -15$$

$$\text{Risk} = \exp(\log\text{-Odds}) / (1 + \exp(\log\text{-Odds}))$$

10 year Heart Failure Risk	
Women	$\begin{aligned} \text{log-Odds} = & -4.310409 + 0.8998235 \times (\text{age} - 55) / 10 - 0.4559771 \times (\text{min}(\text{SBP}, 110) - 110) / 20 \\ & + 0.3576505 \times (\text{max}(\text{SBP}, 110) - 130) / 20 + 1.038346 \times (\text{if diabetes}) + 0.583916 \\ & \times (\text{if current smoker}) - 0.0072294 \times (\text{min}(\text{BMI}, 30) - 25) / 5 + 0.2997706 \times (\text{max}(\text{BMI}, 30) - 30) / 5 \\ & + 0.7451638 \times (\text{min}(\text{eGFR}, 60) - 60) / -15 + 0.0557087 \times (\text{max}(\text{eGFR}, 60) - 90) / -15 \\ & + 0.3534442 \times (\text{if using anti-hypertensive medication}) - 0.0981511 \times (\text{if using anti-hypertensive medication}) \\ & \times (\text{max}(\text{SBP}, 110) - 130) / 20 - 0.0946663 \times (\text{age} - 55) / 10 \times (\text{max}(\text{SBP}, 110) - 130) / 20 \\ & - 0.3581041 \times (\text{age} - 55) / 10 \times (\text{if diabetes}) - 0.1159453 \times (\text{age} - 55) / 10 \times (\text{if current smoker}) \\ & - 0.003878 \times (\text{max}(\text{BMI}, 30) - 30) / 5 - 0.1884289 \times (\text{age} - 55) / 10 \times (\text{min}(\text{eGFR}, 60) - 60) / -15 \end{aligned}$ $\text{Risk} = \exp(\log\text{-Odds}) / (1 + \exp(\log\text{-Odds}))$
Men	$\begin{aligned} \text{log-Odds} = & -3.946391 + 0.8972642 \times (\text{age} - 55) / 10 - 0.6811466 \times (\text{min}(\text{SBP}, 110) - 110) / 20 \\ & + 0.3634461 \times (\text{max}(\text{SBP}, 110) - 130) / 20 + 0.923776 \times (\text{if diabetes}) + 0.5023736 \times (\text{if current smoker}) \\ & - 0.0485841 \times (\text{min}(\text{BMI}, 30) - 25) / 5 + 0.3726929 \times (\text{max}(\text{BMI}, 30) - 30) / 5 \\ & + 0.6926917 \times (\text{min}(\text{eGFR}, 60) - 60) / -15 + 0.0251827 \times (\text{max}(\text{eGFR}, 60) - 90) / -15 \\ & + 0.2980922 \times (\text{if using anti-hypertensive medication}) - 0.0497731 \times (\text{if using anti-hypertensive medication}) \\ & \times (\text{max}(\text{SBP}, 110) - 130) / 20 - 0.1289201 \times (\text{age} - 55) / 10 \times (\text{max}(\text{SBP}, 110) - 130) / 20 \\ & - 0.3040924 \times (\text{age} - 55) / 10 \times (\text{if diabetes}) - 0.1401688 \times (\text{age} - 55) / 10 \times (\text{if current smoker}) \\ & + 0.0068126 \times (\text{max}(\text{BMI}, 30) - 30) / 5 - 0.1797778 \times (\text{age} - 55) / 10 \times (\text{min}(\text{eGFR}, 60) - 60) / -15 \end{aligned}$ $\text{Risk} = \exp(\log\text{-Odds}) / (1 + \exp(\log\text{-Odds}))$

Table S25. Example: 10 year total CVD risk of 50-year-old women, 200 mg/dL Total cholesterol, 45 mg/dL HDL cholesterol, Treated SBP 160 mm Hg, Diabetes, No smoking, eGFR 90 ml/min/1.73m²

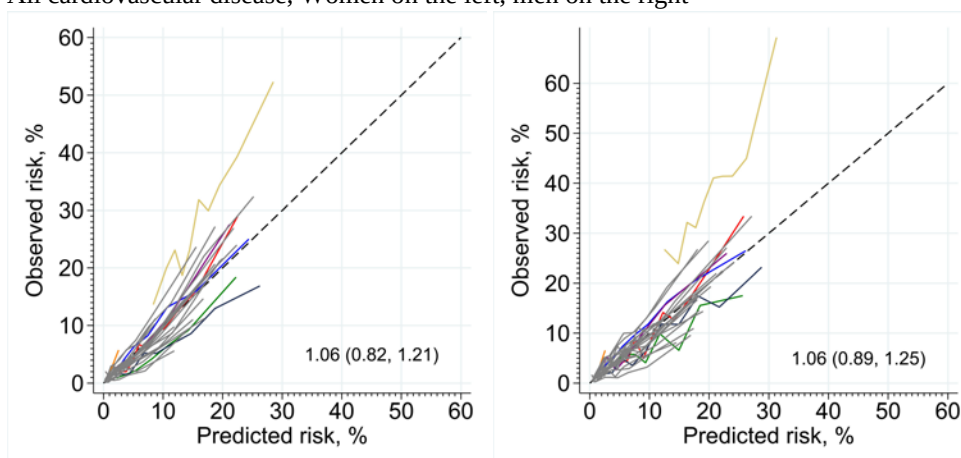
Risk factors	Original input value	Predictor value after required transformation	Coefficient	Product of predictor value and coefficient
Age, year	50	$(50 - 55) / 10 = -0.5$	0.7939329	-0.39697
Total cholesterol, mg/dL*	200	used to calculate non-HDL cholesterol		
non-HDL cholesterol	---	$(200 - 45) \times 0.02586 - 3.5 = 0.5083$	0.0305239	0.015515
HDL cholesterol, mg/dL*	45	$(45 \times 0.02586 - 1.3) / 0.3 = -0.45433$	-0.16069	0.073005
Systolic BP, mmHg	160			
Systolic BP <110, mmHg	---	$(\text{min}(160, 110) - 110) / 20 = 0$	-0.2394	0
Systolic BP 110+, mmHg	---	$(\text{max}(160, 110) - 130) / 20 = 1.5$	0.360078	0.540117
Diabetes mellitus	1	1	0.86676	0.86676
Current smoking	0	0	0.536074	0
eGFR, ml/min/1.73m ²	90			
eGFR <60, ml	---	$(\text{min}(90, 60) - 60) / -15 = 0$	0.604592	0
eGFR 60+, ml	---	$(\text{max}(90, 60) - 90) / -15 = 0$	0.043377	0
HTN meds	1	1	0.315167	0.315167

Statin	0	0	-0.14777	0
Treated SBP 110+	spline	1.5	-0.06636	-0.09954
Treated non-HDLC	spline	0	0.119788	0
Age × non-HDL cholesterol	interaction	$-0.5 \times 0.5083 = -0.25415$	-0.08197	0.020833
Age × HDL cholesterol	interaction	$-0.5 \times -0.45433 = 0.227167$	0.030677	0.006969
Age × Systolic BP 110+	interaction	$-0.5 \times 1.5 = -0.75$	-0.09463	0.070976
Age × Diabetes mellitus	interaction	$-0.5 \times 1 = -0.5$	-0.27057	0.135285
Age × Current smoking	interaction	$-0.5 \times 0 = 0$	-0.07872	0
Age × eGFR <60	interaction	$-0.5 \times 0 = 0$	-0.16378	0
Constant			-3.30773	-3.30773
Sum				-1.75961
Risk = $\exp(-1.75961) / (1 + \exp(-1.75961)) = 0.146839 = 14.68\%$				

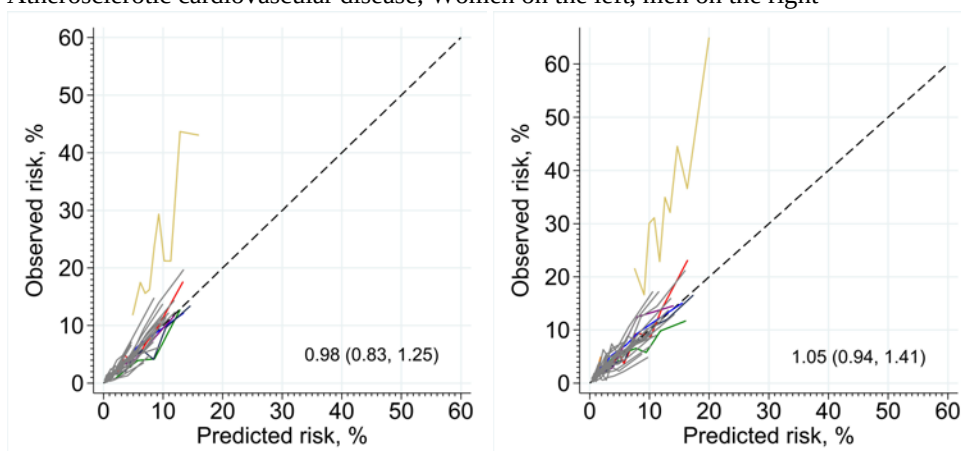
*The equations use cholesterol in mg/dL and apply unit conversion (1 mmol/L = 0.02586 mg/dL)

Supplemental Figure S1. Sex-specific calibration plots in the derivation sample for the PREVENT base model for total CVD, ASCVD and HF.

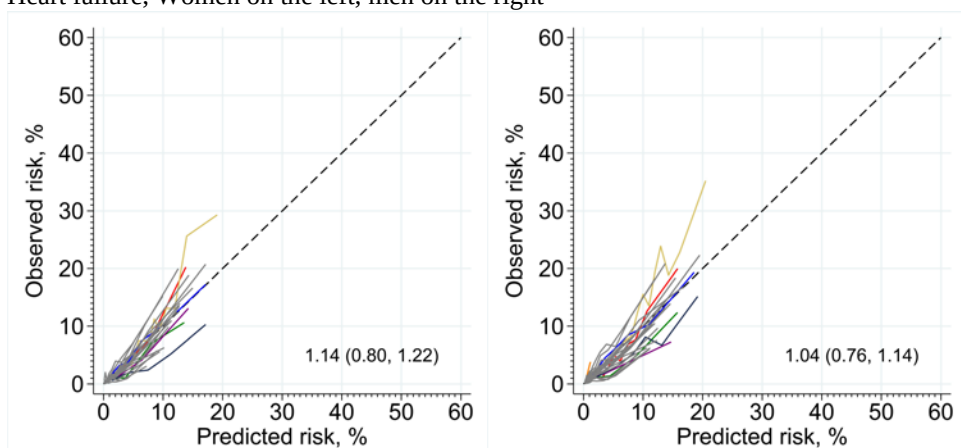
All cardiovascular disease, Women on the left, men on the right



Atherosclerotic cardiovascular disease, Women on the left, men on the right



Heart failure, Women on the left, men on the right



Predicted vs. observed risk by decile within each derivation cohort. Line color key: ARIC red, CARDIA orange, CHS sand, Framingham green, Geisinger blue, JHS purple, MESA dark navy, OLDW cohorts gray

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Appendix 2):

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